SIMATS ENGINEERING

ASSESSMENT TOOL 2

CO3 AT2 Data Interpretation

Name: M Apurva

Reg no: 192325138

Slot: A

Course Code: DSA0404

Course Name: Fundamentals of Data Science

DOS: 17/08/2026

1. A population has mean μ = 500 and SD σ = 80. A random sample of n = 64 is drawn.

Sampling Distribution Reasoning

Given:

- Population mean, $\mu = 500$
- Population standard deviation, $\sigma = 80$
- Sample size, $n = 64$

(a) Compute the standard error of the sample mean.

Step 1:

Use the standard error formula:

$SE = \sigma / \sqrt{n}$

Step 2:

Substitute the given values:

$SE = 80 / \sqrt{64}$

Step 3:

Calculate $\sqrt{64}$:

$\sqrt{64} = 8$

Step 4:

Calculate the standard error:

$SE = 80 / 8 = 10$

$\boxed{\textbf{Final Answer: SE = 10}}$

(b) What is the probability that the sample mean exceeds 520, assuming the sampling distribution is approximately normal?

Step 1:

We need $P(\bar{X} > 520)$.

Step 2:

Calculate the Z-score:

$Z = (\bar{X} - \mu) / SE$

Step 3:

Substitute the values:

$Z = (520 - 500) / 10$

$Z = 20 / 10 = 2$

Step 4:

From the standard normal table:

$P(Z < 2.00) = 0.9772$

Step 5:

We need the area to the right of $Z = 2$:

$P(Z > 2) = 1 - 0.9772$

$P(Z > 2) = 0.0228$

$\boxed{\textbf{Final Answer: } P(\bar{X} > 520) = 0.0228 = 2.28\%}$

(c) If n is increased to 256, how does the standard error change?

Step 1:

New sample size:

$n = 256$

Step 2:

Use the standard error formula:

$SE = \sigma / \sqrt{n}$

Step 3:

Substitute:

$SE = 80 / \sqrt{256}$

Step 4:

Calculate $\sqrt{256}$:

$\sqrt{256} = 16$

Step 5:

Calculate:

SE = 80 / 16 = 5

Final Answer: The standard error decreases from 10 to 5.

Principle: Standard error is inversely proportional to the square root of the sample size. Therefore, increasing the sample size makes the sample mean more precise.

2. Covariance Calculation and Sign Interpretation

X: 10, 20, 30, 40, 50

Y: 15, 18, 25, 30, 42

Calculate Cov(X,Y). Is the relationship direction consistent with intuition? What does the magnitude alone fail to tell you (that correlation would)?

Given:

Advertising Spend (X)	Sales (Y)
10	15
20	18
30	25
40	30
50	42

Step 1:

Calculate the mean of X:

$\bar{X} = (10 + 20 + 30 + 40 + 50) / 5$

$\bar{X} = 150 / 5 = 30$

Step 2:

Calculate the mean of Y:

$\bar{Y} = (15 + 18 + 25 + 30 + 42) / 5$

$\bar{Y} = 130 / 5 = 26$

Step 3:

Use the sample covariance formula:

$Cov(X,Y) = \Sigma[(X_i - \bar{X})(Y_i - \bar{Y})] / (n - 1)$

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	Product
10	15	-20	-11	220
20	18	-10	-8	80
30	25	0	-1	0
40	30	10	4	40
50	42	20	16	320

Step 4:

Add the products:

$\Sigma[(Xi - \bar{X})(Yi - \bar{Y})] = 220 + 80 + 0 + 40 + 320$

$= 660$

Step 5:

Divide by $n - 1$:

$Cov(X,Y) = 660 / (5 - 1)$

$= 660 / 4$

$= 165$

Final Answer: Cov (X, Y) = 165

Interpretation: The covariance is positive. Therefore, advertising spend and sales tend to increase together. This is consistent with intuition.

Important: The magnitude of covariance alone does not tell us the strength of the relationship because covariance depends on the measurement units. Correlation removes the effect of units and gives a standardized value between −1 and +1.

3. Correlation vs Causation Trap

A city reports that ice cream sales and drowning incidents have a correlation coefficient of r = 0.85. A newspaper claims "ice cream causes drowning." Identify the statistical flaw, name the likely confounding variable, and explain what additional evidence (not just correlation) would be needed to claim causation.

Given:

Correlation coefficient, $r = 0.85$

(a) Identify the statistical flaw.

Step 1:

$r = 0.85$ indicates a strong positive correlation.

Step 2:

A strong correlation means that the two variables tend to change together.

Step 3:

It does NOT prove that one variable causes the other.

Final Answer: The newspaper is making the error of confusing correlation with causation.

(b) Identify the likely confounding variable.

The likely confounding variable is temperature or season, especially hot summer weather.

- Hot weather increases ice cream sales.
- Hot weather can increase swimming and outdoor activity.
- Increased swimming activity can increase exposure to drowning incidents.

Therefore, temperature can influence both ice cream sales and drowning incidents.

(c) What additional evidence is needed to claim causation?

Step 1: Researchers would need evidence beyond a correlation coefficient.

Step 2: They should control for confounding factors such as temperature, season, population, and swimming activity.

Step 3: Well-designed longitudinal or experimental evidence should support a causal relationship.

Step 4: The relationship should be supported by a plausible causal mechanism and consistent findings across settings.

Final Answer: Correlation alone cannot establish causation.

4. Effect of Outliers on Correlation

A dataset of 9 points shows r = 0.42 (weak positive). One extreme outlier is removed, and r becomes 0.89. What does this reveal about the fragility of Pearson's correlation? Would Spearman's rank correlation be more robust here? Justify.

Given:

Original correlation: $r = 0.42$

Correlation after removing one extreme outlier: $r = 0.89$

Step 1:

Compare the two correlation values:

$0.42 \rightarrow 0.89$

Step 2:

The correlation has increased substantially after removing only one observation.

Step 3:

This means the extreme outlier had a strong influence on Pearson's correlation.

Step 4:

Therefore, Pearson's correlation can be fragile when extreme observations are present.

Final Answer: The result demonstrates that Pearson's correlation is sensitive to outliers.

Would Spearman's rank correlation be more robust?

Step 1: Spearman's correlation is calculated using the ranks of the observations.

Step 2: Because it uses ranks rather than raw numerical values, extreme numerical values generally have less influence.

Step 3: Therefore, Spearman's rank correlation is generally more robust to outliers.

Final Answer: Yes. Spearman's rank correlation would generally be more robust here, although it is not completely immune to unusual observations.

5. Data Preparation — Missing Values

A dataset of 200 rows has 15% missing values in the "Income" column, and analysis shows these missing values are more common among unemployed respondents (not random). (a) Classify this missing data mechanism (MCAR/MAR/MNAR). (b) Why would mean imputation introduce bias here? (c) Suggest a better strategy.

Given:

- Total number of rows = 200
- Missing Income values = 15%
- Missing values are more common among unemployed respondents.

Step 1:

Calculate the number of missing Income values:

Missing values = 200 × 15 / 100

= 30

Therefore, 30 Income values are missing.

(a) Classify the missing-data mechanism.

Step 1:

The missingness is not completely random because it is more common among unemployed respondents.

Step 2:

Employment status is an observed variable.

Step 3:

Missingness related to an observed variable is classified as MAR under the usual assumption.

Final Answer: MAR — Missing At Random.

Note: If missingness depends on the actual unreported income even after accounting for observed variables, it could be MNAR. Based on the information given, MAR is the expected answer.

(b) Why would mean imputation introduce bias?

Step 1:

Mean imputation replaces every missing Income value with the overall mean income.

Step 2:

The missing values are more common among unemployed respondents.

Step 3:

Unemployed respondents may have income levels that differ from the overall mean.

Step 4:

Replacing their missing values with the overall mean can therefore distort the data.

- It can bias the estimated statistics.
- It reduces the variability of Income.
- It can distort relationships between Income and other variables.
- It ignores the relationship between employment status and Income.

Final Answer: Mean imputation can introduce bias because the missing values are not randomly distributed.

(c) Suggest a better strategy.

Step 1:

Identify variables related to Income and missingness, such as employment status and other relevant demographic or financial variables.

Step 2:

Use these variables to predict plausible Income values.

Step 3:

Prefer multiple imputation or an appropriate model-based imputation method.

Final Answer: Use multiple imputation or predictive/model-based imputation instead of simple mean imputation.

Final Answers

1(a): SE = 10

1(b): $P(\bar{X} > 520) = 0.0228 = 2.28\%$

1(c): SE = 5 when n = 256; SE decreases as sample size increases.

2: Cov(X,Y) = 165; positive covariance indicates a positive direction.

3. Correlation does not imply causation; temperature/season is a likely confounder.

4. Pearson correlation is sensitive to outliers; Spearman is generally more robust.

5(a): MAR

5(b): Mean imputation can introduce bias because missingness is related to employment status.

5(c): Use multiple imputation or model-based imputation.